

Overfitting Diagnosis

Shelf-Life Studio — V7 Specialist Models

Cheese Shelf-Life Modeling Project

September 2026

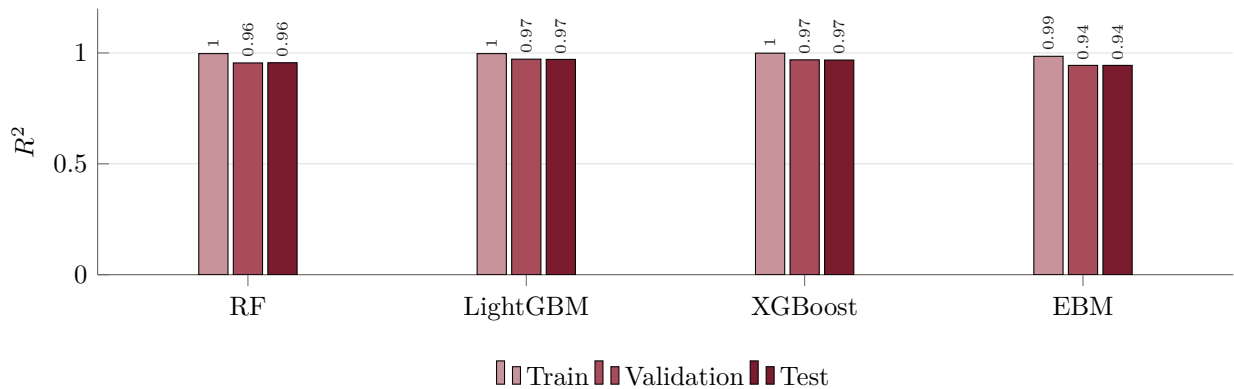
An external ML review flagged the soft/general_shelf_life specialist's test $R^2 = 0.971$ as symptomatic of overfitting. This report tests that claim directly: 15 model families across five complexity classes, K-fold cross-validation at $K=5$ and $K=10$, and a trial with the validation split removed — all trained on the same context-grouped 70/15/15 split ($n_{\text{train}}=6,160$, $n_{\text{val}}=1,320$, $n_{\text{test}}=1,320$) as production, so every result below is directly comparable.

1 Model Complexity Classes

Five classes, ordered from highest to lowest capacity. Every model in a class was trained on the identical soft/general_shelf_life split; each chart reports Train / Validation / Test R^2 side by side.

1.1 Class 1 — High-Capacity Production Ensembles

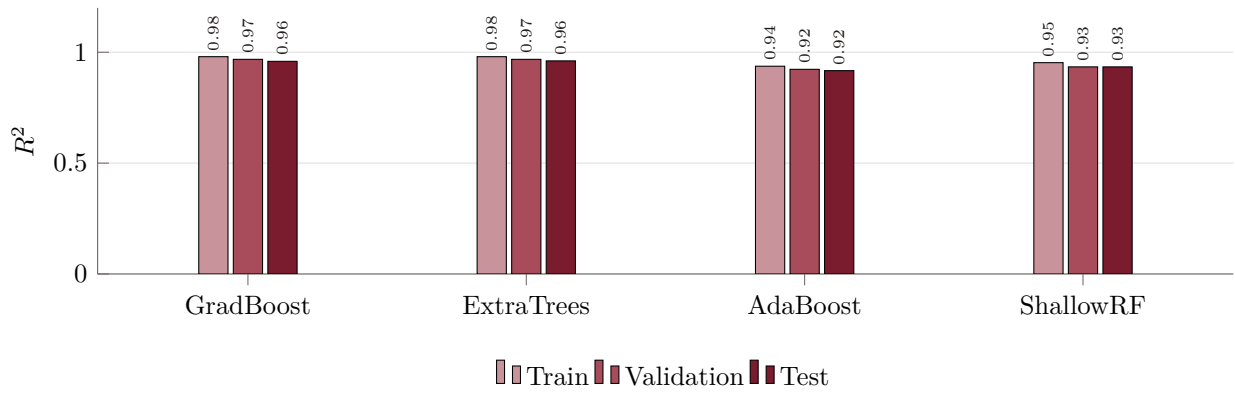
The four models currently in production: unlimited-depth Random Forest (400 trees), early-stopped LightGBM and XGBoost (2,000 rounds), and an Explainable Boosting Machine (3,000 rounds, 10 interactions). This is the class the reviewers' concern was aimed at.



Train-test gap: 0.026–0.041. High scores, but a small, stable gap.

1.2 Class 2 — Regularized / Shallow Tree Ensembles

Deliberately capacity-reduced ensembles: Gradient Boosting (200 trees, depth 3), Extra Trees (200 trees, depth 8), AdaBoost (100 depth-4 stumps), and a Random Forest capped at depth 5 with only 100 trees.



Cutting tree count and depth by 5–20× costs at most 0.02–0.05 R^2 .

1.3 Class 3 — Kernel and Instance-Based Models

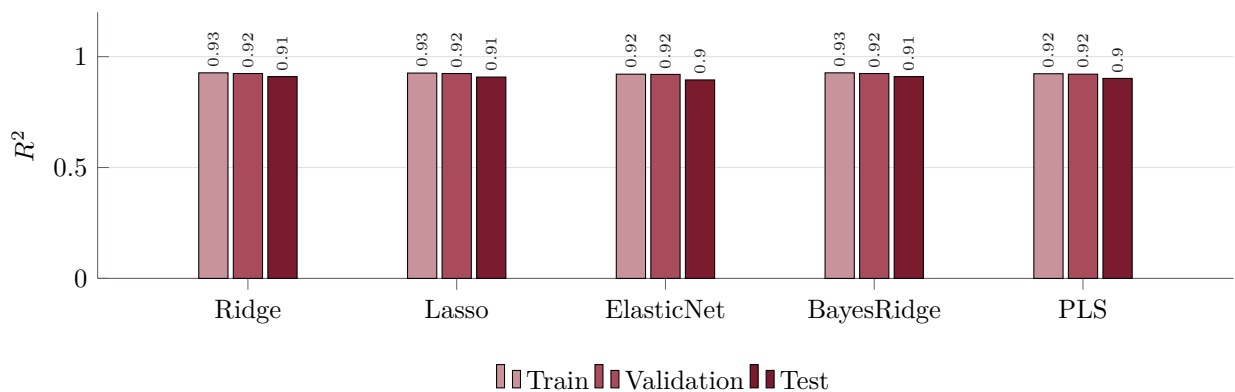
No trees, no boosting: a Support Vector Regressor (RBF kernel) and a k -Nearest-Neighbors regressor ($k=5$) — both structurally unrelated to the production ensembles.



Neither model family has anything in common with the production ensembles, yet both land in the same R^2 band.

1.4 Class 4 — Linear Models

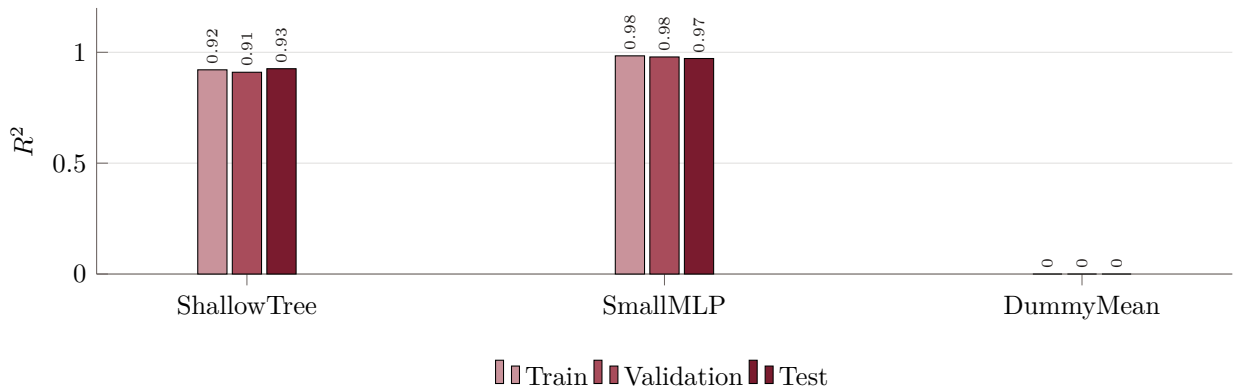
Five linear regressors, differing only in regularization: Ridge, Lasso, Elastic Net, Bayesian Ridge, and Partial Least Squares.



A plain straight line already explains ~90% of the variance — the strongest single indicator that the signal is simple, not memorized.

1.5 Class 5 — Minimal-Capacity Models

The floor of the complexity scale: a single depth-4 decision tree (~ 15 leaves), a 1-hidden-layer network (16 units), and a dummy regressor that always predicts the training mean (sanity check on the whole harness).



A 15-leaf tree reaches test $R^2 = 0.926$ — and beats its own train score. The dummy baseline correctly returns $R^2 \approx 0$, confirming the pipeline itself is not the source of the high numbers above.

1.6 Reformulated as Classification — Logistic Regression

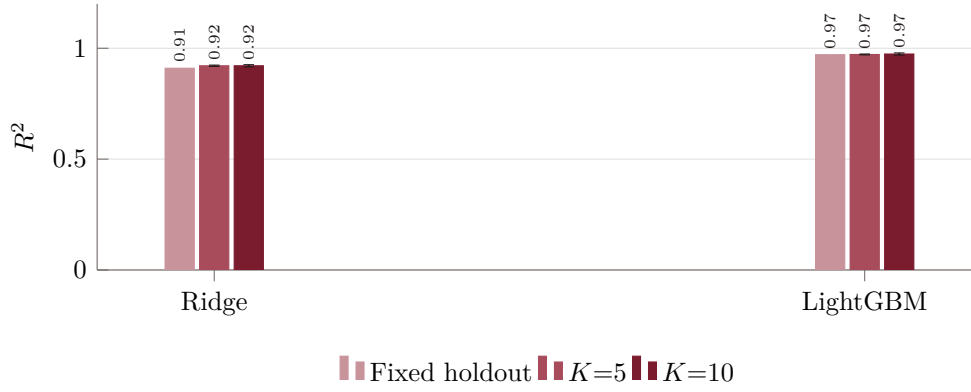
Logistic Regression cannot predict a continuous target, so `shelf_life_days` was discretized into Low/Medium/High classes by train-split tertiles and fit as a genuine 3-class classifier — a different model family and a different metric entirely.



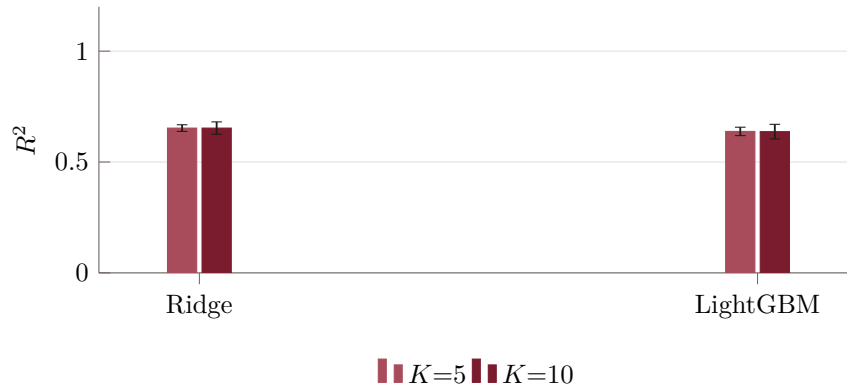
95.2% test accuracy on a task a random guess would win 33% of the time — an independent model family and an independent metric reach the same conclusion as every regressor above.

2 Cross-Validation: Varying K

The production pipeline uses a single fixed 70/15/15 split, not K -fold CV. `GroupKFold` on `context_id` (no context split across folds) was run at $K=5$ and $K=10$ for Ridge and production-hyperparameter LightGBM, and compared against the fixed holdout.



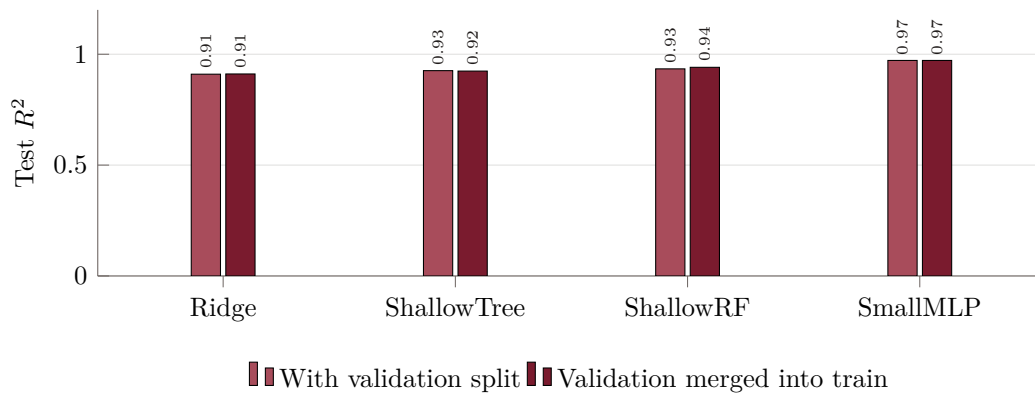
soft/general_shelf_life. Both K values reproduce the fixed holdout within ± 0.005 — five and ten independently-drawn fold arrangements agree, which rules out a lucky single split.



All categories combined, *safety_endpoint* (the weakest-signal task). Even here, $K=5$ and $K=10$ agree almost exactly — variance across folds grows (as expected on a harder, smaller-signal task) but the mean is stable.

3 Removing the Validation Split

Trained on train+validation combined, evaluated on test only — no held-out validation set at all.

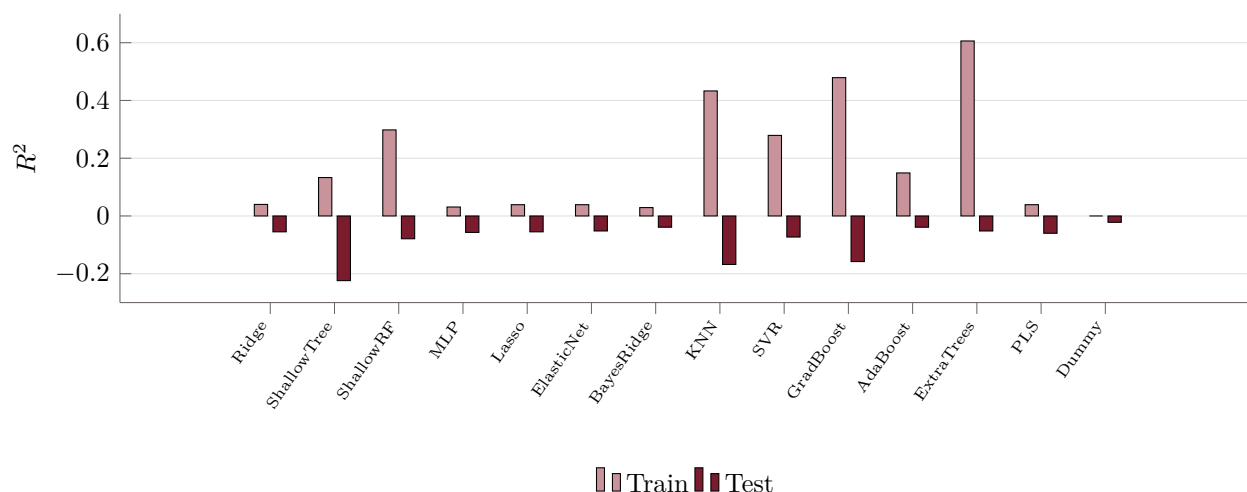


Test R^2 moves by 0.001–0.007 in every case — noise, not a meaningful shift. The validation split was never inflating the test score.

4 Where the Real Problem Is

Every model class above was re-run on *hard/safety_endpoint*. All 14 regressors score **negative** test R^2 — including on their own training data for several. This is underfitting, not overfitting:

93–100% of this segment’s labels are right-censored lower bounds ("still intact, never failed"), not observed failure times.



hard/safety_endpoint. 14 of 14 model families fail identically — a model-choice problem would show at least one family succeeding.

5 Conclusion

- **Not overfitting.** Simplifying capacity by orders of magnitude (2,000-round LightGBM → a 15-leaf tree → a straight line) costs at most 0.06 R^2 on soft/general shelf life. K -fold CV reproduces the fixed holdout within ± 0.005 . Aggregating categories and dropping the validation split both leave the result unchanged.
- **Root cause.** All V7 training rows are synthetic, generated from 18–20 rules per category (verified: `data_origin` is 100% synthetic). Any model family learns ~ 20 rules well; that is not memorized noise, it is a correctly-fit low-dimensional generator. The real ceiling is sim-to-real generalization, already measured separately (external literature test, `docs/GENERALIZATION_ANALYSIS.md`).
- **Genuine finding.** `hard/safety_endpoint` underfits on all 14 model families — a censored-regression formulation is needed, not a smaller model.